

SYNENT Technology Internship

Task 3: To-Do List

1. Introduction

The To-Do List is a command-line Python application developed as part of the SYNENT Technology Internship. It allows users to add, view, and delete tasks through a simple menu-driven interface.

2. Objective

The project demonstrates Python lists, functions, loops, conditional statements, user input, validation, and exception handling.

3. Technologies Used

Technology	Purpose
Python 3	Programming language used to build the application
List	Stores the user's tasks
CLI / Terminal	Provides interaction and displays results

4. Methodology

- Tasks are stored in a Python list.
- A menu-driven while loop keeps the application running.
- Functions separately handle adding, viewing, and deleting tasks.
- Tasks are displayed with numbers for easier selection.
- Input is validated to prevent empty tasks and invalid task numbers.
- try-except handles non-numeric task-number input.

5. Code Review and Improvements

The initial uploaded version provided add, delete, view, and exit options. The reviewed version improves maintainability by using functions, safer deletion by task number, input validation, clearer prompts, numbered task output, and a main() entry point.

6. Features

• Add tasks • View tasks • Delete tasks • Empty-input prevention • Invalid-input handling • Clean menu interface

7. Sample Output

```
TO-DO LIST
```

1. Add Task
2. Delete Task
3. View Tasks
4. Exit

Enter your choice (1-4): 1
Enter your task: Complete Python assignment
Task added successfully!

8. Result

The application successfully manages a list of tasks through a simple command-line interface and handles common invalid inputs safely.

9. Conclusion

This project provides practical experience with Python data structures, functions, loops, conditional logic, validation, and exception handling.